

ARTICLE



Fecal microbiota transplantation in capsules for the treatment of steroid refractory and steroid dependent acute graft vs. host disease: a pilot study

Ilan Youngster^{1,2,11}, Adi Eshel^{3,11}, Mika Geva⁴, Ivetta Danylesko⁵, Israel Henig⁵, Tsila Zuckerman^{5,6}, Shalev Fried^{2,4}, Ronit Yerushalmi^{2,4}, Noga Shem-Tov^{2,4}, Joshua A. Fein⁷, David Bomze^{6,8}, Avichai Shimoni^{2,4}, Omry Koren^{3,12}, Roni Shouval^{2,4,9,10,12} and Arnon Nagler^{2,4,12}

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Acute graft-versus-host disease (aGvHD) is a serious complication of allogeneic hematopoietic stem-cell transplantation with limited treatment options. The gut microbiome plays a critical role in aGvHD pathogenesis. Fecal microbiota transplantation (FMT) has emerged as a potential therapeutic approach to restore gut microbial diversity. In this prospective pilot study, 21 patients with steroid-resistant or steroid-dependent lower gastrointestinal aGvHD received FMT in capsule form. At 28 days after the first FMT, the overall response rate was 52.4%, with 23.8% complete and 28.6% partial responses. However, sustained responses were infrequent, with only one patient remaining aGvHD-free long-term. FMT was generally well-tolerated. Microbiome analysis revealed dysbiosis in pre-FMT patient stool samples, with distinct microbial characteristics compared to donors. Following FMT, there was an increase in beneficial Clostridiales and a decrease in pathogenic Enterobacteriales. These findings highlight the potential of FMT as a treatment option for steroid-resistant aGvHD. Trial registration number NCT #03214289.

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INTRODUCTION

Acute graft-versus-host disease (aGvHD) is a common and potentially fatal complication of allogeneic hematopoietic stem-cell transplantation (HCT), caused by activated, donor-derived T cells attacking host tissue [1]. Immense efforts notwithstanding, drivers of GvHD have not been completely understood and the overall incidence and resulting mortality is still high, despite recent introduction of novel immune-modulatory treatment modalities [2]. Corticosteroids are the first line of therapy for aGvHD, but their administration results in complete response rates of approximately 50% [3]. For patients who don't respond to steroids, ruxolitinib is a viable option. However, durable responses to ruxolitinib only occur in 40% of patients [2]. Thus, definitive treatment for aGvHD remains an unmet need.

While the pathophysiology of GvHD is not completely understood, studies have shown that the gut microbiome is a central feature and driver of the disease [4–8]. Several authors have reported an association between low gut microbiota diversity and increased mortality in adult and pediatric allogeneic HCT recipients. Reduced amounts of anti-inflammatory bacteria (e.g. *Faecalibacterium prausnitzii*) and over-domination by other bacteria (e.g., *Enterococcus*) are linked to severe presentations of gastrointestinal (GI) GvHD. During HCT patients are exposed to

conditioning regimens, antibiotics and altered diets, which all have a profound effect on the gut microbiome, often leading to severe microbial dysbiosis [9, 10].

Fecal microbiota transplantation (FMT), the installation of stool from a healthy donor to the GI tract of a patient, is increasingly used as a treatment modality to restore a dysbiotic microbiome. At present FMT is a commonly accepted treatment for recurrent *Clostridioides difficile* infection, however numerous studies are investigating FMT for other indications. As aGvHD is believed to be, at least partially, related to gut microbiome perturbations, FMT could potentially be of benefit to these patients. To date, several case reports and relatively small case series have reported FMT to be safe and effective for patients with steroid refractory (SR) aGvHD [11], however data collected in a prospective standardized manner is sorely needed.

In the current study we aimed to evaluate the safety and efficacy of FMT for the treatment of steroid resistant and steroid dependent GI aGvHD.

METHODS

Patients and study design

This was an extension of a single-arm, non-randomized pilot study (NCT #03214289) evaluating the safety and efficacy of FMT for patients with

¹Shamir Medical Center, Beer Yaacov, Israel. ²Faculty of Medicine, Tel Aviv University, Tel Aviv, Israel. ³Azrieli Faculty of Medicine, Bar Ilan University, Safed, Israel. ⁴Hematology and Bone Marrow Transplantation Division, Chaim Sheba Medical Center, Tel Hashomer, Tel Aviv, Israel. ⁵Department of Hematology and Bone Marrow Transplantation, Rambam Health Care Campus, Haifa, Israel. ⁶Ruth and Bruce Rappaport Faculty of Medicine, Technion, Haifa, Israel. ⁷Department of Medicine, Weill Cornell Medical College, New York, NY, USA. ⁸Division of Dermatology, Sourasky Medical Center, Tel Aviv, Israel. ⁹Adult BMT Service, Memorial Sloan Kettering Cancer Center, New York, NY, USA. ¹⁰Weill Cornell Medical College, New York, NY, USA. ¹¹These authors contributed equally: Ilan Youngster, Adi Eshel. ¹²These authors jointly supervised this work: Omry Koren, Roni Shouval, Arnon Nagler. ✉email: shouvalr@mskcc.org

Table 1. Population characteristics.

Age (median [IQR])		51.6 [40.0, 64.5]
Sex (%)	Female	10 (47.6)
	Male	11 (52.4)
Diagnosis (%)	Acute lymphoblastic leukemia	1 (4.8)
	Acute myeloid leukemia	7 (33.3)
	Lymphoma	3 (14.3)
	Myelodysplastic syndrome	6 (28.6)
	Multiple myeloma	1 (4.8)
	Myeloproliferative neoplasms	3 (14.3)
Remission at time of HCT	No	11 (52.4)
	Yes	10 (47.6)
HCT donor (%)	Haploidentical	1 (4.8)
	HLA- Mismatched unrelated 9/10	3 (14.3)
	HLA- Matched sibling donor	10 (47.6)
	HLA- Matched unrelated donor	7 (33.3)
Myeloablative conditioning (%)	No	8 (38.1)
	Yes	13 (61.9)
In-vivo T-cell depletion with antithymocyte globulin (Neovii) (%)	No	7 (33.3)
	Yes	14 (66.7)
GvHD prophylaxis ^a (%)	MMF	8 (38.1)
	MMF & PTCY	1 (4.8)
	MTX	10 (47.6)
	MTX & MMF	2 (9.5)
Days from HCT to GvHD onset (median [IQR])		90 [35, 147]
Biopsy proven GvHD (%)	No	10 (47.6)
	Yes	11 (52.4)
Gastrointestinal GvHD stage at onset (%)	0	0 (0)
	1	0 (0)
	2	1 (4.8)
	3	17 (81.0)
	4	3 (14.3)
Skin GvHD stage at onset (%)	0	16 (76.2)
	1	1 (4.8)
	2	3 (14.3)
	3	1 (4.8)
	4	0 (0)
Liver GvHD stage at onset (%)	0	17 (81.0)
	1	2 (9.5)
	2	1 (4.8)
	3	1 (4.8)
	4	0 (0)
IBMTR grade at onset (%)	A	0 (0)
	B	0 (0)
	C	14 (66.7)
	D	7 (33.3)

Table 1. continued

Total parenteral nutrition (%) before first FMT	No	13 (61.9)
	Yes	8 (38.1)
Methylprednisolone mg/kg at time for first FMT (median [IQR])		1.6 [1.0, 2.0]
GvHD treatment before FMT		
Cyclosporine (%)	No	5 (23.8)
	Yes	16 (76.2)
MMF (%)	No	2 (9.5)
	Yes	19 (90.5)
Budesonide (%)	Yes	21 (100.0)
Mesenchymal stem cells (%)	No	19 (90.5)
	Yes	2 (9.5)
Vedolizumab (%)	No	19 (90.5)
	Yes	2 (9.5)
Extracorporeal photopheresis (%)	No	16 (76.2)
	Yes	5 (23.8)
Ruxolitinib (%)	No	17 (81.0)
	Yes	4 (19.0)
Days from GvHD to first FMT (median [IQR])		15 [10, 28]

IQR Interquartile range, HCT Hematopoietic cell transplantation, MMF Mycophenolate mofetil, MTX Methotrexate, PTCY Post-transplantation, IBMTR International Bone Marrow Transplant Registry, GvHD Graft versus host disease, FMT Fecal microbiota transplantation.

^aGiven with a calcineurin inhibitor.

steroid-resistant or steroid-dependent lower gastrointestinal (LGI) aGvHD. Twenty-one patients were enrolled from two centers (Chaim Sheba Medical Center, Tel Hashomer, Ramat Gan, Israel [$n = 17$] and Rambam Health Care Campus, Haifa, Israel [$n = 4$]).

Patients included met the following criteria: 1) Age ≥ 16 years and ≤ 75 years; 2) Active LGI aGvHD diagnosis, confirmed by standard clinical criteria or biopsy [12–14]; 3) Meeting criteria for steroid-resistant or steroid-dependent LGI aGvHD. Steroid-resistant LGI aGvHD was defined as intestinal GVHD that did not improve within seven days after initial steroid therapy (≥ 1 mg/kg of methylprednisolone) or progressed after five days of treatment. Steroid-dependent LGI aGvHD was defined as recurrence of LGI aGvHD during steroid tapering [15].

Response criteria to treatment follows standard definitions (see Supplementary Materials) [16, 17].

The study was approved by institutional review boards and all patients signed informed consent before enrollment.

The first four patients enrolled were treated under a clinical trial NCT03214289. The remaining patients were continued on the same protocol and were treated in a prospective manner.

Donor recruitment and FMT procedure

Donors were unrelated healthy, non-pregnant adults aged 18–50 years, taking no medications, with normal body mass index. Volunteers were excluded for any significant medical history or the use of any antibiotics in the six months preceding stool collection. Candidates were screened with the American Association of Blood Banks donor questionnaire and underwent a physical examination, and general laboratory screening tests as described previously, in accordance with the guidance of the Israeli Ministry of Health for FMT [18]. Additional testing for capsules administered in this immunocompromised population included polymerase chain reaction testing for cytomegalovirus and screening of stool aliquots for vancomycin-resistant enterococci, methicillin-resistant staphylococcus aureus, and carbapenemase-resistant Enterobacteriaceae (CRE).

Capsules were produced at the Center for Microbiome Research at Shamir Medical Center, Israel, following a previously described protocol [19] and stored frozen at -80°C before use. Further information can be found in the supplementary materials.

Participants received an oral FMT course: 15 capsules daily for two consecutive days (30 capsules total). In patients with no or partial response, FMT could be repeated from the same or a different donor at the treating physician's discretion. Participants were asked to fast for 4 h before and 1 h following capsule intake. Capsules were taken with sips of water. There was a recommendation to stop any antibiotic treatment 48 h before FMT administration. Subjects were followed for at least six months post-FMT.

Bacterial DNA extraction, 16S amplification, and sequencing

Bacterial DNA was extracted from 100 mg of stool, amplified by PCR (V4 region of the bacterial 16S gene), and sequenced (Illumina MiSeq) as described in the supplementary materials.

16S rRNA gene sequence analysis

Data analysis utilized Quantitative Insights Into Microbial Ecology software (QIIME2) [20]. Paired-end sequences were demultiplexed using per-sample barcodes, then error-corrected through Divisive Amplicon Denoising Algorithm (DADA2) [21]. Feature sequences were aligned against GreenGenes database [22] with 99% confidence for taxonomic annotation. Low abundance features (<0.001%), mitochondria and chloroplast sequences were filtered out.

RESULTS

Population characteristics

Baseline characteristics of 21 adult patients with a median age of 51 years (interquartile range (IQR) 40–64.5) treated with FMT are listed in Table 1. Acute myeloid leukemia was the leading diagnosis (33.3%). Allogeneic transplants were performed between March 2017 to February 2019 and were primarily from matched sibling donors (47.6%) and matched unrelated donors (33.3%). Myeloablative conditioning regimens were used in 61.9% of patients. Cyclosporine was uniformly administered as the backbone for GvHD prophylaxis; Anti-thymocyte globulin was used in 66.7% of cases.

The median time from HCT to aGvHD was 90 days (IQR: 35, 147), with all subjects developing at least stage 2 LGI aGvHD and grade C-D aGvHD. Skin and liver were involved in the minority of cases

(23.8% and 19%, respectively). Only one patient had steroid-dependent aGvHD, with the remaining being steroid-resistant. At time of study entry, the median number of GvHD-directed treatment lines was 2 (IQR 2–3), with only one patient receiving one treatment line before FMT. Before the first FMT, the median prednisone-equivalent dose was 1.6 mg/kg (IQR 1.0–2.0). Other GvHD-directed drugs included budesonide (100%), mycophenolate mofetil (90.5%), cyclosporine (76.2%), and ruxolitinib (19.0%). Additional earlier interventions for aGvHD included extracorporeal photopheresis (23.8%), vedolizumab (9.5%), mesenchymal stem cell (9.5%).

Efficacy

The median time from aGvHD onset to the first FMT was 15 days (IQR 10–28; Fig. 1). A total of 43 FMT courses from 7 unique donors were administered. Of the 21 patients in the study, 6 (28.6%) received a single FMT course, 8 (38.1%) a second course, and 7 (33.3%) a third course. The FMT donor was changed in 9 of the 22 repeated FMTs.

At 28 days after the first FMT, the overall response rate (ORR) was 52.4% (complete response (CR) 23.8%, partial response (PR) 28.6%). ORR measured 28 days after the last FMT was 25% (CR 25%, PR 0%). Considering all courses of FMT, the best response was 42.9% CR and 18% PR (Fig. 2a). The median duration of response was 44 days (IQR 11–56).

Median survival was 52 days (95% CI 36–103). Survival was numerically higher among patients achieving a best response of CR or PR (i.e., responders) compared to non-responders – median survival 57 days (95% CI 40 – non calculable) vs. 35 days (95% CI 35 – non calculable); $p = 0.12$ (Fig. 2b). Only one patient remained alive and aGvHD free with long-term follow-up (414 days from first FMT). AGvHD, often resulting in sepsis, was the primary cause of death (14/20; 70%). Five patients (25%) died without active GvHD; causes of death included (infections [$n = 2$], relapse [$n = 2$], and a neurological disorder [$n = 1$]). In one patient (5%) cause of death was unknown. Of note, infection was the leading or contributing cause of death in 17 cases. Among the 8 patients who achieved a

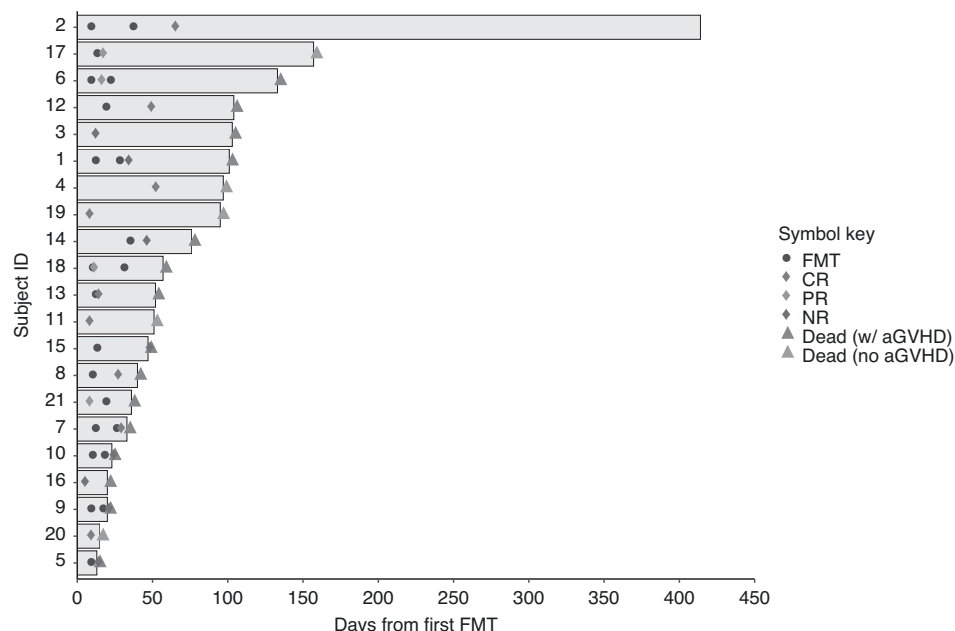


Fig. 1 Patient trajectories following FMT. This swimmer plot delineates the dynamic clinical trajectories experienced by each patient after FMT. Each horizontal line corresponds to an individual patient, and the x-axis represents time in days since the initiation of the first FMT course (day zero). Red circles signify subsequent FMT courses, while diamond shapes denote key response assessment points. Complete responses (CR) are indicated by blue, partial responses (PR) by orange and no responses (NR) by green. Triangles mark patient deaths, with purple denoting death with aGvHD and blue signifying death without aGvHD.

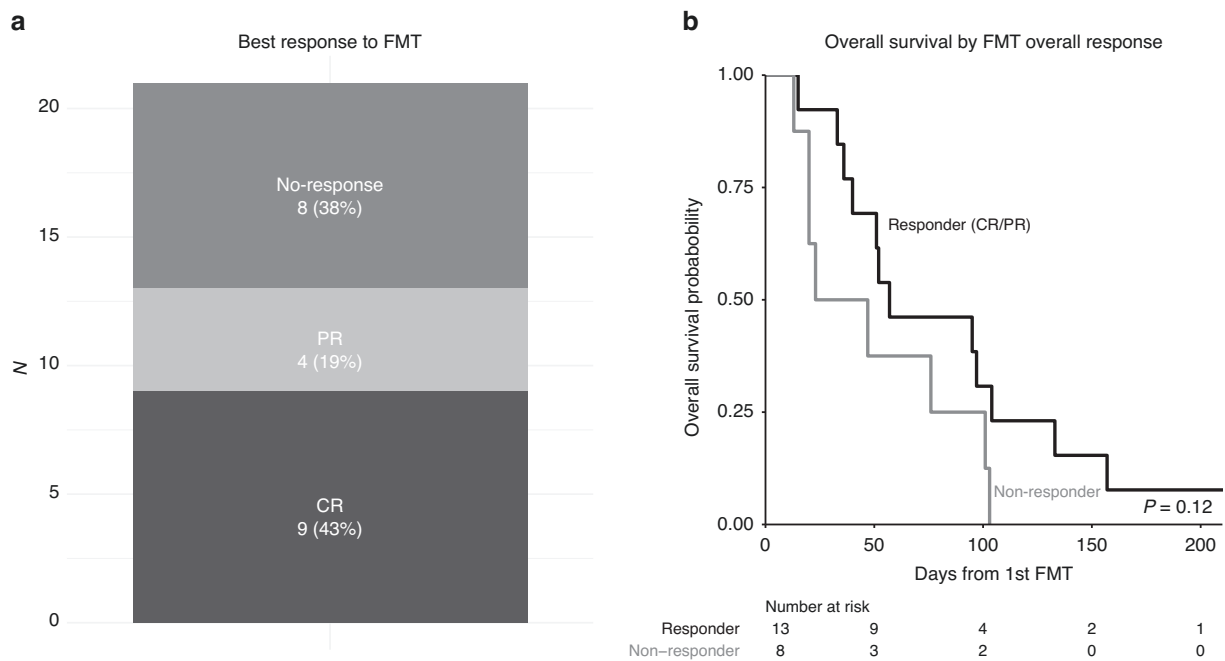


Fig. 2 Response and overall survival following FMT. The impact of FMT according to patients' response. **a** Patients' best response following FMT, the y axis represents the number of patients according to each of the responses – complete response (CR), partial response (PR). **b** Kaplan–Meier estimates of overall survival by best response to FMT. Comparison was performed with a log-rank test.

Table 2. Adverse events after FMT.

Adverse event	n
Bacteremia	33
Viral infections	10
GI bleeding	8
Fungal infection	3
Nausea	2
Diarrhea	1
Heart failure	1
Pneumonia	1
Pulmonary hemorrhage	1
UTI	1
Vomiting	1

GI Gastrointestinal (GI), UTI Urinary tract infections.

best response of CR and subsequently died, aGvHD was the main cause of death (3/8), followed by infection (3/8) and disease relapse (2/8).

In patients receiving one FMT vs. two or more, CR rates were (4/6; 67%) and (5/15; 33%), respectively. Corresponding median overall survival from day of first FMT was 73 days (95% CI 20 – non calculable) and 52 days (95% CI 36–133).

Safety

Adverse events following FMT are described in Table 2. Overall, FMT in capsules was well tolerated. Two events of nausea and one of vomiting occurred shortly following FMT. Nine events of GI bleeding (primarily rectal bleeding $n = 7$) occurred at a median of 10 days (IQR 7–12, range 1–24) after the last course of FMT. These events occurred in the setting of thrombocytopenia and severe GI GvHD.

A total of 33 blood stream infection (BSI) events (Supplementary Table S1) occurred at a median of 7 days (IQR 4–19; range 1–25) after the last FMT. *Enterococcus faecium* ($n = 10$), *Escherichia coli* ($n = 5$), *Klebsiella pneumoniae* ($n = 3$), and *Pseudomonas aeruginosa* ($n = 3$) were the most common etiologies. As previously reported, we performed metagenomic sequencing on the FMT donations, stool samples from the patients, and bacterial strains isolated from the blood cultures of 13 BSI cases [23]. In all BSI instances, the offending strains were not identified in the FMT stool donations. However, in 5 cases the BSI-inciting bacteria were identified in pre-FMT patient stool samples. Other infectious adverse events (AEs) included viral infections ($n = 10$), which were primarily cytomegalovirus (CMV) reactivations, fungal infections ($n = 3$), urinary tract infection ($n = 1$), and a pneumonia ($n = 1$).

Antibacterial antibiotics were frequently used after FMT. In 25/43 (58%) of FMT procedures, antibiotics were started within a median of 5 days; the counting of days was relative to the FMT conducted closest to antibiotic initiation and preceding it. Piperacillin-tazobactam (40%), quinolones (28%), and vancomycin (16%) were the most frequently prescribed antibiotics. Among the instances where antibiotics were administered, a discernible bacterial source was identified in 14 out of 25 cases.

Overall, aside from nausea and vomiting, investigators deemed AEs to be not directly related to FMT.

Patients with SR GvHD have a dysbiotic microbiome

Patients' stool samples collected before the first FMT were dysbiotic, having lower α -diversity (measured by the Shannon Index) compared to FMT donors' stool samples (Fig. 3a). β -diversity distances between samples were calculated using Weighted UniFrac. Pre-FMT stool samples were significantly dissimilar to donor samples (Fig. 3b) and showed significant higher distances within the group, indicating distinct microbial characteristics in pre-FMT patients (Fig. 3c). Taxonomic composition in donor samples reflected abundance of expected bacterial commensal in a healthy population. Pre-FMT patient samples were enriched for pathobionts such as unclassified members of

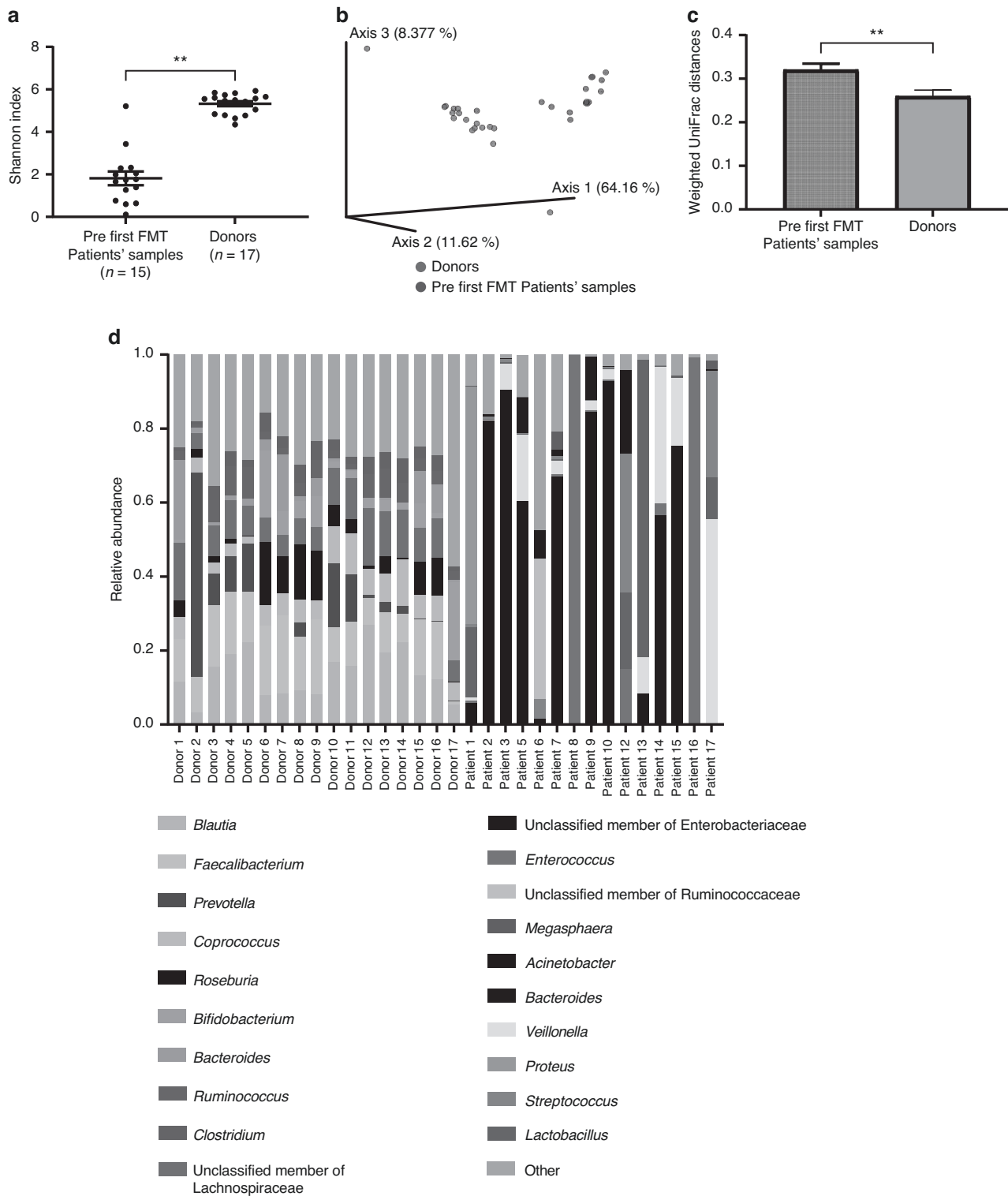


Fig. 3 Microbiota characterization of pre-FMT patients' samples. **a** Shannon index comparison between patients' stool samples collected before the first FMT session and donors' stool samples. (** P value < 0.001 Mann–Witney test). **b** Principal Coordinates Analysis (PCoA) based on Weighted UniFrac distance matrix of FMT donors (red) and pre-FMT patients' stool samples. $q = 0.001$ for the distances between the groups. **c** Weighted UniFrac distances within each study – Patients' stool samples collected before the first FMT session and FMT donors' samples. (** P value < 0.001 Mann–Witney test). **d** Individual sample's bacteria composition - demonstrated by the relative abundance of the ten most abundant bacteria within each group – pre-first FMT session patients' (right) and donors' samples (left). The relative abundance of other bacteria is marked as "other" in gray.

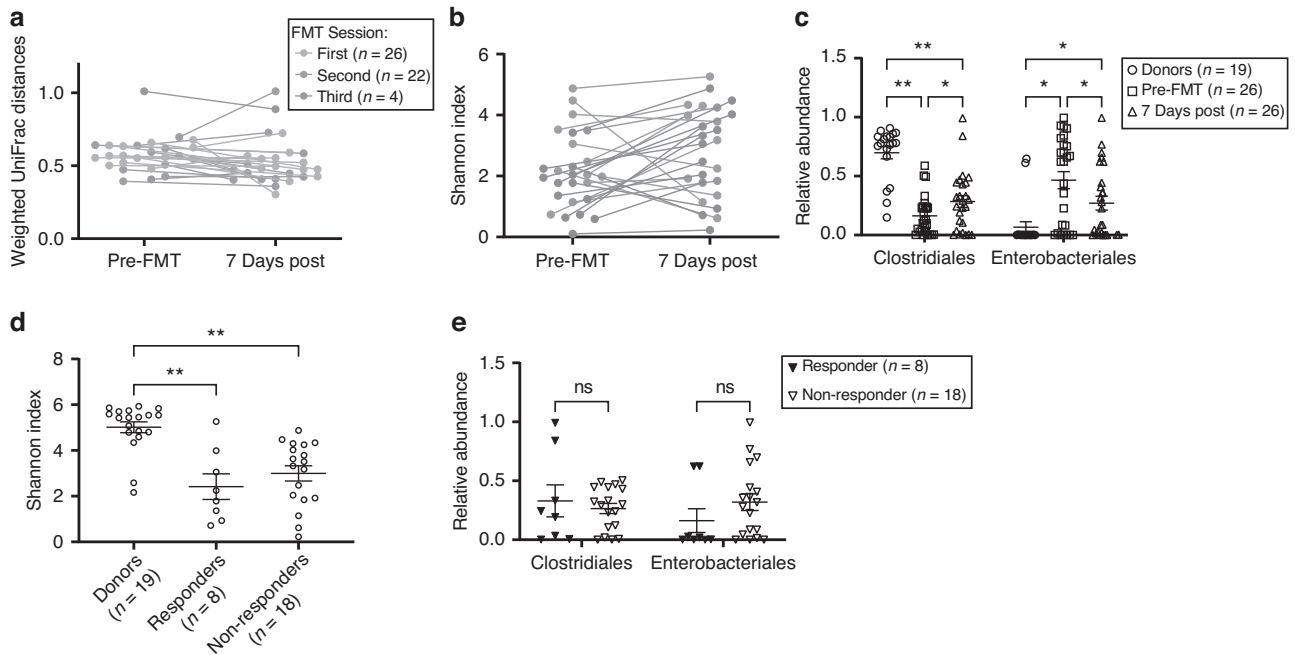


Fig. 4 Microbiota dynamics as a response to FMT treatment. **a** Weighted UniFrac distance between patients and their FMT donors at two different time points – before FMT and seven days after (+–3 days). Each FMT session was calculated separately and marked with different colors (first FMT – green, second FMT – pink, third FMT –blue). The lines connect between pre and post FMT samples from the same individual at a single FMT session. (p -value = 0.01 Wilcoxon signed rank (paired)). **b** Shannon index values at pre and post FMT. Each FMT session was calculated separately and marked with different colors (first FMT – green, second FMT – pink, third FMT –blue). The lines connect between pre and post FMT samples from the same individual at a single FMT session. (p -value = 0.03122 Wilcoxon signed rank (paired)). **c** Comparison of clostridiales and enterobacteriales relative abundance between FMT donors (circle), patients' samples collected before FMT (square), and seven days after (+–3 days) (triangle). (* P value < 0.05, ** P value < 0.001 Mixed effect model). **d** Shannon index in stool samples of FMT donors, patients who exhibit complete or partial response 7 days post the FMT (responders), and patients who did not respond (non-responders). The response rate was estimated after each FMT session separately. (** P value < 0.001 Mann–Witney test). **e** Comparison of clostridiales and enterobacteriales relative abundance between patients who exhibit complete or partial response seven days post the FMT (responders) and patients who did not respond (non-responders). The response rate was estimated after each FMT session separately. (ns not significant Mann–Witney test).

Enterobacteriaceae (Fig. 3d). Following FMT there was considerable inter-patient variability in changes and α - and β -diversity (Fig. 4a, b). Nonetheless, we observed a considerable increase in Clostridiales and a decrease in Enterobacteriales in stool samples collected 7 days after FMT compared to pre-FMT samples (Fig. 4c). There was no difference in alpha-diversity or relative abundance of Clostridiales and Enterobacteriales between responders and non-responders (Fig. 4d, e).

DISCUSSION

In this study, we conducted an assessment of the efficacy and safety of FMT in capsules for patients with SR aGvHD. Approximately 50% of the patients exhibited a response to FMT, with 25% achieving a complete response. Nevertheless, the responses were not sustained, as only one out of 22 patients remained alive and free from aGvHD with long-term follow-up. FMT was well-tolerated; however, a notable increase in BSIs was observed, though a direct causal relationship to the procedure was not firmly established. As expected, patients with GI aGvHD demonstrated a dysbiotic microbiome. Notably, FMT led to a significant rise in Clostridiales and a decline in Enterobacteriales. Overall, our findings suggest that FMT shows promise as a treatment option for patients with SR aGvHD.

FMT has emerged as a potential therapeutic intervention for patients with SR aGvHD following HCT, with several studies showing promising results [24–28]. Kakihana et al. conducted a

pioneering study using fresh FMT from relative donors, and their findings indicated that FMT correlated with increased regulatory T-cells and beneficial bacterial species in patients with GI aGvHD [25]. Similarly, Spindelboeck et al. reported clinical responses in patients with grade IV GvHD after repeated FMTs via colonoscopy [27]. Qi et al. also achieved symptomatic remission in patients with stage IV GvHD using FMT from unrelated donors [28]. Additionally, van Lier et al. demonstrated complete responses in GvHD patients after FMT from unrelated donors via nasoduodenal infusion [26]. A recent meta-analysis of 23 studies found that FMT is effective in treating steroid resistant and steroid dependent GI aGvHD. The pooled clinical remission rate was 64% in prospective single-arm studies and 81% in retrospective studies, case series, and case reports [11]. These studies collectively suggest that FMT holds promise as a viable approach for managing GI aGvHD. Our findings are in line with these response rates, but importantly, our study is unique in allowing for repeated administration of FMT and involved the use of capsules for administration.

FMT has demonstrated a favorable safety profile in various immunosuppressive states [29, 30]. This is particularly relevant for patients with SR GvHD, who are profoundly immunosuppressed. Our study also found FMT to be well-tolerated overall, but we observed high rates of BSIs. In a subset of patients, where the blood culture bacterial isolate was available as well as patient and donor stool samples, we performed a metagenomic investigation and did not identify the source of bacteremia in the FMT donors [23]. Furthermore, repeated infections are expected in SR aGvHD

patients. Nevertheless, FMT remains a concern in this population due to reports of FMT-induced bacteremia in the literature [31]. An indirect involvement of FMT in propagating bacterial translocation from the gut through the damaged intestinal barrier, possibly through a local gut mucosal inflammatory reaction, cannot be ruled out. Therefore, caution should be exercised when performing FMT in this vulnerable population.

It is noteworthy that regardless of response to FMT, infection was a leading or contributing cause of death in the majority patients. This again highlights, the profound susceptibility to infections within this cohort, attributable in part to the heightened state of immunosuppression integral to GVHD treatment. The high degree of immunosuppression appears to play a pivotal role in amplifying the risk of infections, ultimately contributing significantly to patient mortality. Such insights underscore the urgent need for interventions introduced earlier in the course of GVHD that pivot away from heavy reliance on immunosuppressive regimens. Non-immunosuppressive approaches, particularly those centered around microbiota-based interventions, emerge as compelling alternatives. These interventions hold promise in addressing the underlying dynamics of GVHD while mitigating the heightened susceptibility to infections. Given the challenges associated with managing infections in this profoundly immunocompromised patient population, interventions that act independently of traditional immunosuppressive strategies offer a hopeful avenue for enhancing patient outcomes.

Perturbations in the microbiome are characteristic of HCT and, specifically, GvHD [4–6]. Moreover, the microbiome has been identified as a driver of GvHD [5–7]. This has served as the underlying rationale for investigating FMT in GvHD. In previous FMT trials for GvHD intervention, the response to FMT was accompanied by an increase in gut microbial α -diversity, partial engraftment of donor bacterial species, and increased abundance of butyrate-producing bacteria, including species from the Clostridiales and Blautia genera [26, 28]. Following FMT, we observed an increase in Clostridiales, but did not identify a significant correlation with response. Nonetheless, it is important to consider that the study might have been underpowered. Collectively, FMT appears to induce beneficial changes to the microbiome. However, it remains unclear whether these changes translate into a beneficial clinical effect.

The study has several limitations. While sample size is larger than in previous FMT studies, it remains relatively limited, and the cohort is heterogeneous. Additionally, lack of a control group, variability in donors, and variations in timing and dosage of FMT make it challenging to clearly determine beneficial effects of FMT in this difficult-to-treat population. Nevertheless, we did identify a favorable signal, motivating further prospective studies. Administration of FMT in SR-aGVHD is particularly challenging, as many patients are severely cytopenic, prone to infections, and at risk of bleeding. Efficacy of FMT may have been affected by frequent administration of antibiotics in this population. Considering earlier FMT administration upon diagnosis of aGVHD as a preventive strategy may overcome some challenges [32, 33]. Lastly, FMT is not a targeted approach, and there is donor heterogeneity. Use of pooled fecal microbiota and specific microbiota consortia show promise as approaches to harness microbiota-based interventions for improving GVHD treatment [34, 35].

Overall, while FMT shows promise as a therapeutic avenue, further research is needed to optimize its efficacy and safety for the benefit of patients battling this challenging condition. By continuing to investigate the complex interactions between the microbiome and GvHD, we can pave the way for more targeted and effective treatments in the future.

DATA AVAILABILITY

Data are available upon request from the corresponding author and approval of data transfer from the IRB.

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AUTHOR CONTRIBUTIONS

IY, RS, AN planned the study. AE, RS, DB, and RS analyzed the data. RS, ID, and TZ enrolled patients. IY manufactured the fecal microbiota transplant capsules. MG coded data into the database. RS wrote the manuscript. IY, OK, RS, and AN supervised the analysis and study. All authors edited and reviewed the manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to Roni Shouval.

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